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An Artificial Neural Network Based Anomaly Detection Algorithm for Nuclear Power Plants

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INTRODUCTION

Larger cybersecurity surfaces have been created due to the integration of advanced digital technologies in the nuclear power plant (NPP) for improving its performance [1]. Although critical assets, which interconnect plant systems to perform safety, security, and emergency preparedness functions, are designed to be isolated from the internet, the increasing frequency, scale, and sophistication of cyber threats lead to higher probability of penetration of the NPP operation and control system than ever before. In certain scenarios, the attacker, rather than stealing data, may conduct data or signal manipulation aiming to mislead NPP operators to perform corrective actions which may lead to a disastrous outcome. Therefore, **it is desired to develop an automated and robust anomaly detection algorithm to assist plant operators to identify the attack and signal false, localize the anomaly, and minimizing the rate of false response when attacks are in progress.**

Various types of artificial neural networks (ANNs) have been applied to the nuclear power plant monitoring and diagnosis [2-4], but primarily focusing on the failure identification and classification of transient types in early stage and diagnosis before significant accident from the viewpoint of the operational safety and maintenance costs. In those applications, the NPP monitoring and diagnostics models were developed based on a single general purpose ANN or a combination of multiple ANNs for improved robustness and selectivity of the system [5-7]. **The data used for training the network were usually generated from NPP simulators due to the unavailability of the actual plant operation data.**

The objective of this study, as the first attempt of constructing a false signal detection and accommodation system for NPPs, is to develop an ANN-based anomaly detection algorithm, which is formulated using NPP operating data associated with various components collected from a calibrated NPP simulation model, to identify the potential data manipulation as a result of cyber-attacks or signal false in a timely manner and provide early warning to the NPP operators. More specifically, effort is made to determine an optimized ANN with regard to its type and configuration for detecting the anomaly of plant signals under normal operating conditions.

METHODOLOGY

The anomaly detection (also outlier detection) is a technique used to identify patterns in a given dataset that do not conform to an established normal behavior. In this study, the dependency or correlation of NPP state variables for a given normal procedure will be created using a wide range of simulation results and the anomaly detection algorithm will be formulated based on a type of ANN.

Operating Data Collection

The Generic Pressurized Water Reactor (GPWR) simulator [8], which was developed based upon a reference US nuclear plant that has been in operation and training for over 20 years, is utilized to generate the operating data necessary for developing the predictive model for the anomaly detection. The GPWR reference simulator has been built and tested to the American Nuclear Society ANSI/ANS-3.5 Nuclear Simulator Standard [9]. It includes high fidelity models that allow full plant operation including normal operations, abnormal operations and emergency operations as required by ANS-3.5 and the simulator response has been validated against actual operating plant data. This three-loop plant has three steam generators, three reactor coolant pumps, and a pressurizer in the primary system.

As a case study, one of the most basic procedures of the reactor control, the power escalation from 55% to 100% reactor power has been performed on the GPWR simulator to collect the operating data. Since several plant components are involved in the operation, including reactor core, steam generators, pressurizer, turbine, etc., **a wide range of plant-wise state variables with the total number of 81 are recorded over the operation period of 10291 seconds at the rate of 1 measurement per second.**

Table 1 shows a partial list of state variables collected and processed and will be used in the next step. The time-dependent behavior of three variables, including electric power, boron concentration, and the position of control rod bank D, are depicted in Fig. 1 as an example.

Table 1. Example of state variables utilized in ANN training

State Variable	Unit
Gross electric power	MW
Percent full power	%
Pressurizer pressure	MPa
Primary hot/cold leg temperature (Loop 1-3)	°C

Steam generator (SG) pressure (#1-3)	MPa
Main feedwater flow (Loop 1-3)	kg/s
Steam generator steam flow (#1-3)	kg/s
Control rod bank position (A-D)	step
Steam (turbine) load	MW
Pressurizer level	%
Governor valve opening (1-4)	%
Steam generator level (#1-3)	%
Total neutron flux	n/cm ² -s

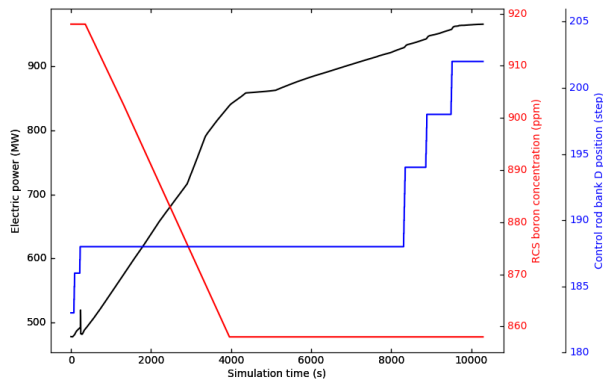


Fig. 1. Selected state variable evolution in power escalation simulation.

Anomaly Detection Procedure

Machine learning techniques have been witnessing increasing roles in the development of anomaly detection algorithms to address the weaknesses of knowledge base detection techniques. The ANN is considered an effective means in this field thanks to its capability of determining non-linear descriptions from the input space into the output space, as well as dependencies between several variables. Neural network consists of simple processing units called neurons. An ANN, in general, has three layers, namely, the input, output and (at least one) hidden layers. Training the ANN is a process of calibrating the synaptic weights (strength between neurons) by performing forward- and back-propagation.

In this study, the auto-associative neural network (AANN) is selected for developing the anomaly detection algorithm where the input and output are the same set of time-dependent state variables (e.g., those listed in Table 1), since the intended application is to identify potential single or multiple signals that have been manipulated due to cyber-attacks or signal false. Fig. 2 depicts the general procedure of the anomaly detection, including the data generation, training of the AANN, and the detection (validation) of abnormal signals.

As mentioned previously, the AANN training data is created for the plant power escalation from 55% to 100% nominal power, which contains $N = 81$ time series x_n ($n = 1, 2, \dots, N$) of length $T = 10291$. At each time step, all state variables, x_1 through x_{81} , formulate one set of data and in total

10291 sets of data are employed to train the AANN. Back-propagation using a sigmoid function is used in the training process. It should be noted that the same dataset is used as input (x_n) and output (y_n) vectors to generate dependencies (correlation) among all state variables. The training process is completed when acceptable learning accomplishes. Under normal operating conditions, it is expected the instantaneous values generated by the trained network are identical to the actual output. In the scenario of a cyber-attack, one or more signals collected would differ from the normal value (x'_n) and consequently introduce deviations to the actual value (y'_n), which is served as the AANN input, while the corresponding predicted value (AANN output) remains virtually constant owing to the influence of all other input variables through the connecting weights. Therefore, the deviation of the actual values in the system (input) from the estimates predicted by the AANN (output) indicates the potential of data manipulation, and to which variable this action may be applied.

In this study, Keras, a Python-based open source deep learning library was used to construct the AANN [10,11]. The SciKit-Learn module was employed to randomly shuffle the dataset for training, testing, and validating the AANN.

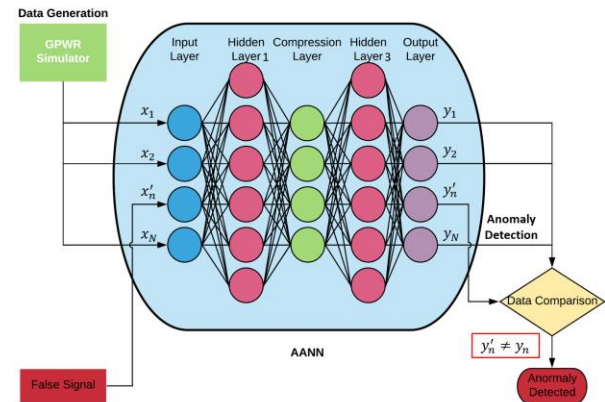


Fig. 2 Procedure of AANN training and testing for anomaly detection.

The simulation data was normalized before being imported to the neural network, because the heterogeneity (range of values) existed in the dataset would trigger a large learning gradient update and preventing the network from converging. More specifically, Eq.(1) was used to normalize each of the N parameters independently to ensure the mean and standard deviation of the time series is 0 and 1, respectively.

$$\hat{x}_n = \frac{x_n - \bar{x}_n}{\sigma(x_n)} \quad (1)$$

where x_n is the input matrix for the n^{th} simulation variable, and $\bar{x}_n$ and $\sigma(x_n)$ is the mean and standard deviation of x_n with along the T time steps.

The AANN with the architecture shown in Fig. 2 was developed in this study. It has five layers, including three non-linear hidden layers between the linear input and output layers. In these layers, the linear input and output layers allow only linear transformation of the data, while hidden layers allow neural network to learn more complicated features. The first and third hidden layers have more units (neurons) than the rest layers. The middle layer, called the compression layer, is included to reinforce neural pattern learning.

The number of neurons was determined by the trail-and-error approach to maximize the performance. And the final architecture contains N , $4N$, N , $4N$ and N neurons in the five layers, respectively.

For each of the N state variables, 70% of the simulation results (or 7204 sets) was selected for the training and the rest (or 3087 sets) was used for the testing. The data selection and random shuffling of their sequence were done using SciKit-Learn. Upon completion of the network training and testing, a new set of data called validation data was created by manually altering one or more state variables in the test set, which would be used to measure the selectivity and robustness of the anomaly detection algorithm.

In order to quantify the performance of the current network, the specific metrics called mean squared error (MSE) which measures the square value of the difference between input and output by the trained network was used in this study. Without going into great details of explanation of this metrics due to the constraint of space, a relatively high accuracy means that output data prediction based on input data is close to input itself and vice versa.

RESULTS

A parametric study was first performed to determine the optimal learning epochs, which is the number of time dataset is used to update the synaptic weight of each neuron, by training the AANN with increasing epochs up to 5000. The trained network corresponding to each epoch was tested using the testing dataset, and the predictive accuracy is depicted in Fig. 3. It can be seen that the accuracy of AANN for predicting both datasets stops improving after ~1000 epochs and the maximum accuracy is 88% at 3000 epochs, indicating that further increase in epochs starts overfitting training data. The accuracy obtained from applying AANN to testing set is only slightly lower than that of the training set, showing the reliability of this AANN when applied to new data.

As a result, 3000 was selected as the optimal number of epochs, based on which an AANN was developed, and it is stored in HDF5 format, along with AANN trained with 500, 1000, 2000 number of epochs, for the validation and eventually the future application for the anomaly detection.

As previously mentioned, the validation set was created by manually altering one or more state variables in the test set. In this study, four validation sets were created by separately manipulating percent full power, control rod bank

position D, SG channel level, and primary loop temperature, while their time-dependent values were replaced by associated mean values calculated over the entire simulation.

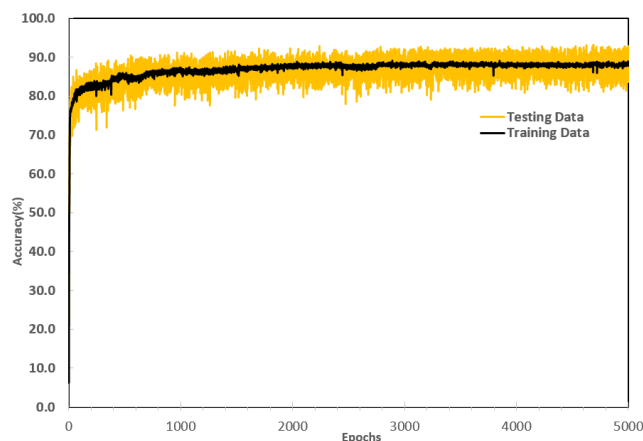


Fig. 3 The evolving training and testing accuracy of AANN.

Since an ideal neural network would reproduce the input variable in the output matrix, by feeding the neural network with the manipulated validation data, any significant decrease in prediction accuracy of the validation set can be used as an indication of ongoing anomalous behaviors in the NPP system.

Following detection algorithm above, 4 sets of AANN models that previously identified and preserved were used to evaluate the performance of the anomaly detection algorithm, as plotted in Fig. 4. The black straight line shows the training accuracy of each model, and blue bars represent the prediction accuracy of the corresponding validation set using AANN trained. A significant decrease, in the range of 10 – 25%, in prediction accuracy of each state variable has been observed in all 4 models, which shows that the trained AANN models are sensitive to the anomalous behavior of even only one state variable.

It should also be noted that, in this case, although 3000 is the optimal number of epochs for AANN training, the behavior of AANN that trained with 500, 1000 and 2000 epochs also show somewhat high-level of sensitivity to the false signal and thus can be used to identify the anomalies in NPP.

In order to visualize the prediction result obtained by the well-trained AANN model, another two NPP responses (SG #1 channel level and SG #1 pressure) were selected due to their significant fluctuation during simulation. As depicted in Fig. 5, each of these two responses was replaced by the mean of the actual data and used as input in validation calculation. The predicted data were plotted in comparison to original data, which shows good agreement over the entire simulation. The maximum error was found in the prediction of SG channel level at around 300 s to 600 s, which is still as small as less than 1.50%.

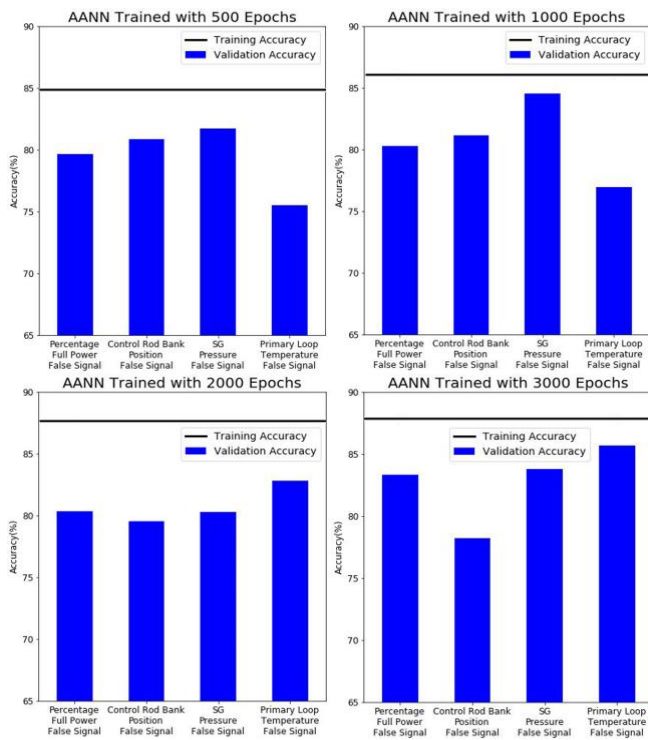


Fig. 4 Predictive accuracy corresponding to selected manipulated parameters using four different AANN models.

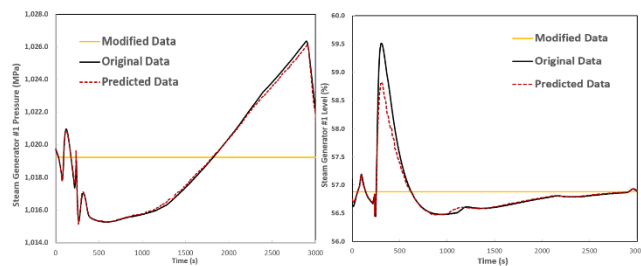


Fig. 5 Comparison between predicted data, original data, and modified data to selected manipulated parameters using well-trained AANN model.

CONCLUSION

An ANN-based anomaly detection algorithm has been developed to detect abnormal behaviors perceivable on the instrument panel of the NPP due to either sensor malfunction or cyber-attack. The simulation data generated using the GPWR simulator for normal power maneuvering operations were utilized to construct the AANN, aiming to reveal the underlying correlations of the state variables in the time series. Preliminary results show that the well-trained AANN can, with satisfactory accuracy, detect the anomaly when the value of only one of the state variables is manually altered and thus indicates it can be potentially used to monitor the operation condition of the NPP and provide crucial assistance to the operator for the system diagnosis.

The future study will focus on refining the AANN performance by improving its predictive capability. In addition, varying degree of data alternation will be carried out to investigate the selectivity and robustness of the anomaly detection algorithm. And the analog and digital alarms on the control panel will be included as part of the input and output vectors in addition to aforementioned state variables. It is also planned to extend the current algorithm developed for the normal operations to transients, where the main focus is devoted to the anomaly detection before and after the onset of a fast transient in order to develop a correct knowledge of the plant conditions. Moreover, component-wise anomaly detection models, in contrast to the plant-wise models shown in this study, will be developed and investigated for the potential improvement of the detection accuracy for local events that do not have an impact across the entire NPP system.

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